Julius-Maximilians-
UNIVERSITÄT
WÜRZBURG

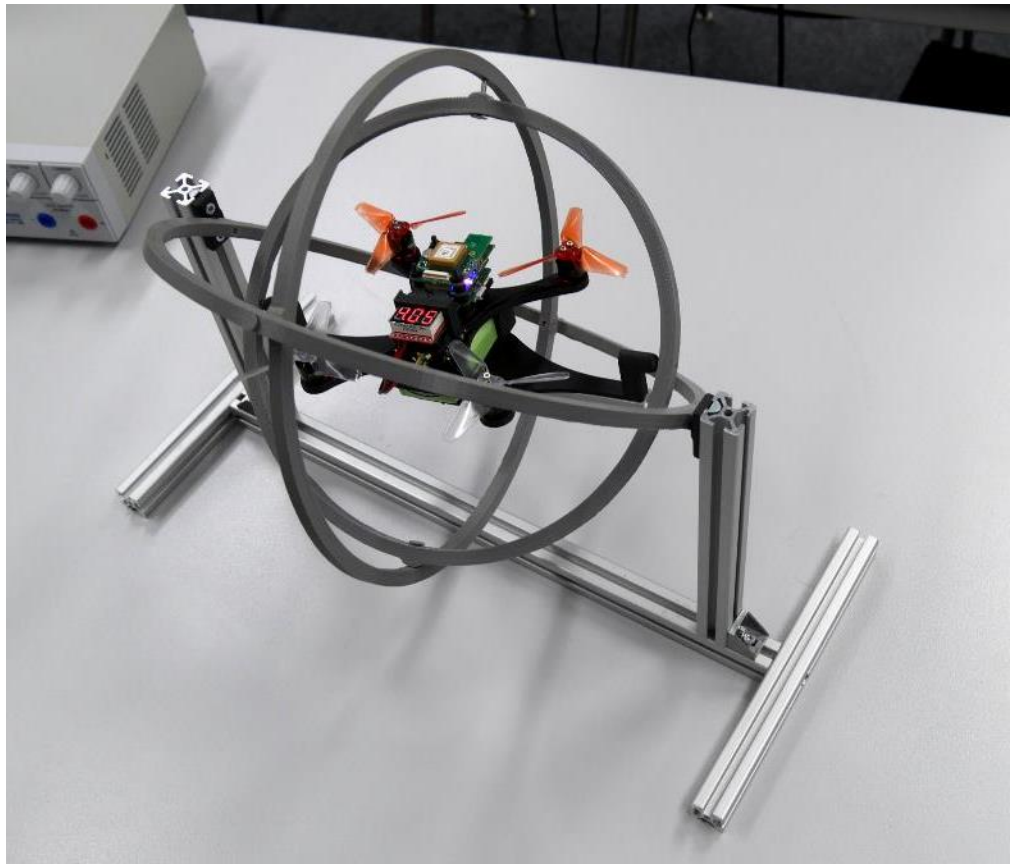
Chair of Aerospace Information Technology

Online Floatsat 2021
Attitude Determination – Part 1

Julian Rothe

Introduction

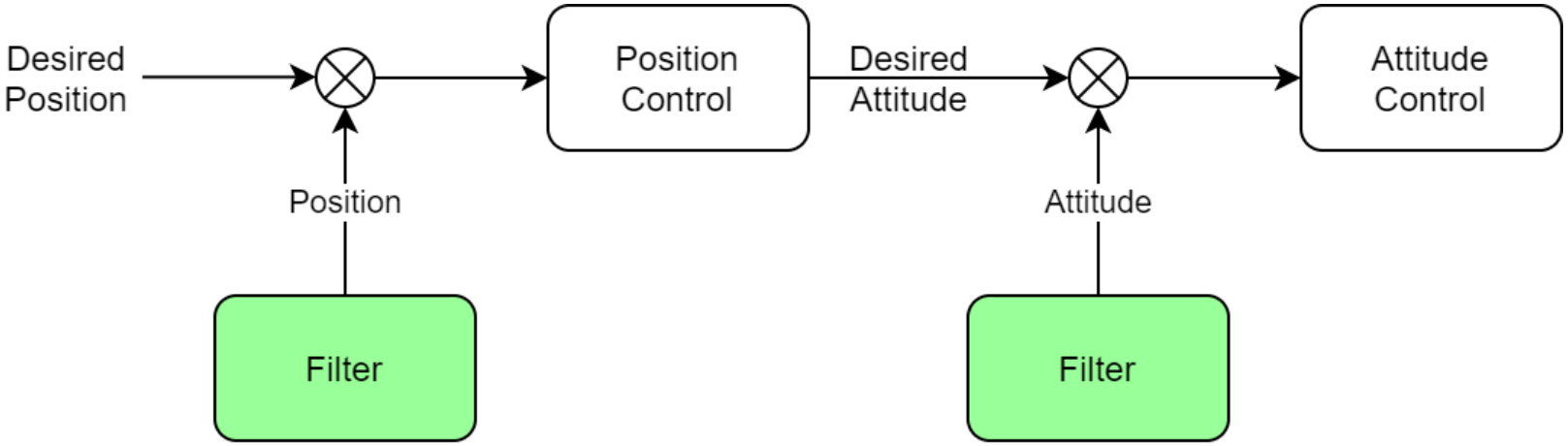
Introduction



Introduction

Why is a Filter so important?

The Controller is highly dependend on a good Filter!



What are we going to learn?

Part1: Simple Complementary Filter

- IMU Basics
- Determine the attitude for each sensor in our IMU

  → Gyroscope, Accelerometer and Magnetometer
- Combine these sensors into a simple AHRS with a complementary filter
- Implement your own embedded complementary filter

What are we going to learn?

Part2: Problems, Quaternions, Matlib and Quaternion Filter

- Problems in the reality
- What are Quaternions, why do we need them?
- Improve our previous complementary filter by using quaternions
- RODOS Matlib – This will make things much easier!
- Implement an embedded quaternion attitude filter

What are we going to learn?

Part3: Complex Filter Design and Implementation

- Understanding and implementing more complex filters

→ Mahony Filter and Madgwick Filter

IMU Basics

- IMU = Inertial Measurement Unit
- Consists of (one of) the following sensors:
  - Gyroscope → Angular Velocity ω
  - Accelerometer → Linear Acceleration a
  - Magnetometer → Magnetic Field m
- In the exercises we use all of the sensors above

IMU Basics

- A combination of Gyroscope and Accelerometer

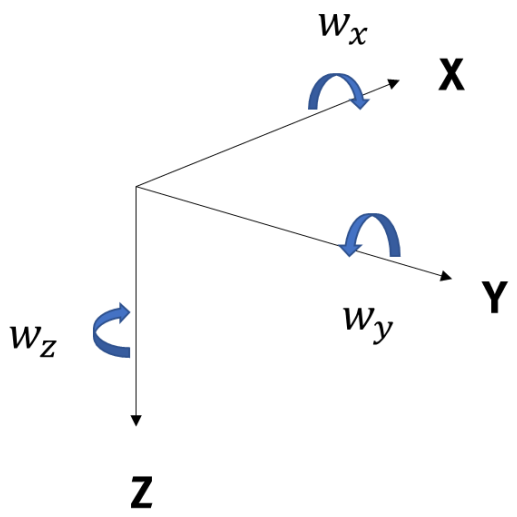
  → 6-DoF IMU

- A combination of Gyroscope, Accelerometer and Magnetometer

  → 9-Dof IMU or AHRS (Attitude and Heading Reference System)

IMU Basics

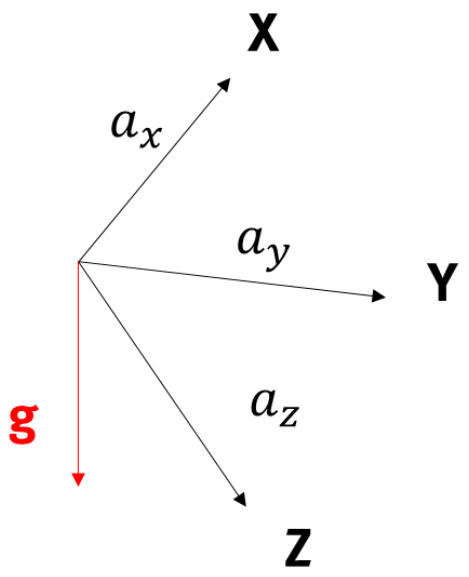
Gyroscope measures the angular velocity in 3 axis



$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T$$

IMU Basics

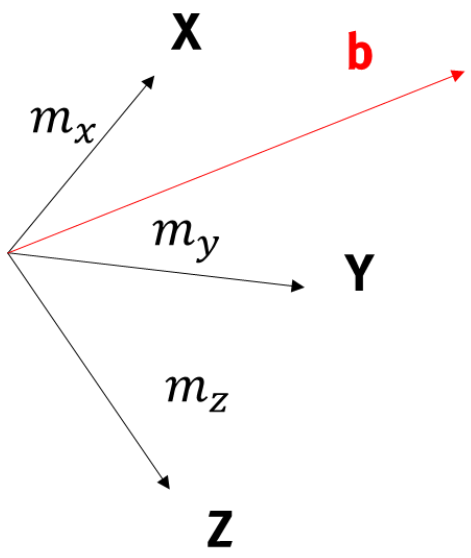
Accelerometer measures the linear acceleration in 3 axis



$$a = \left[a_x \; a_y \; a_z\right]^T$$

IMU Basics

Magnetometer measures the magnetic field in 3 axis

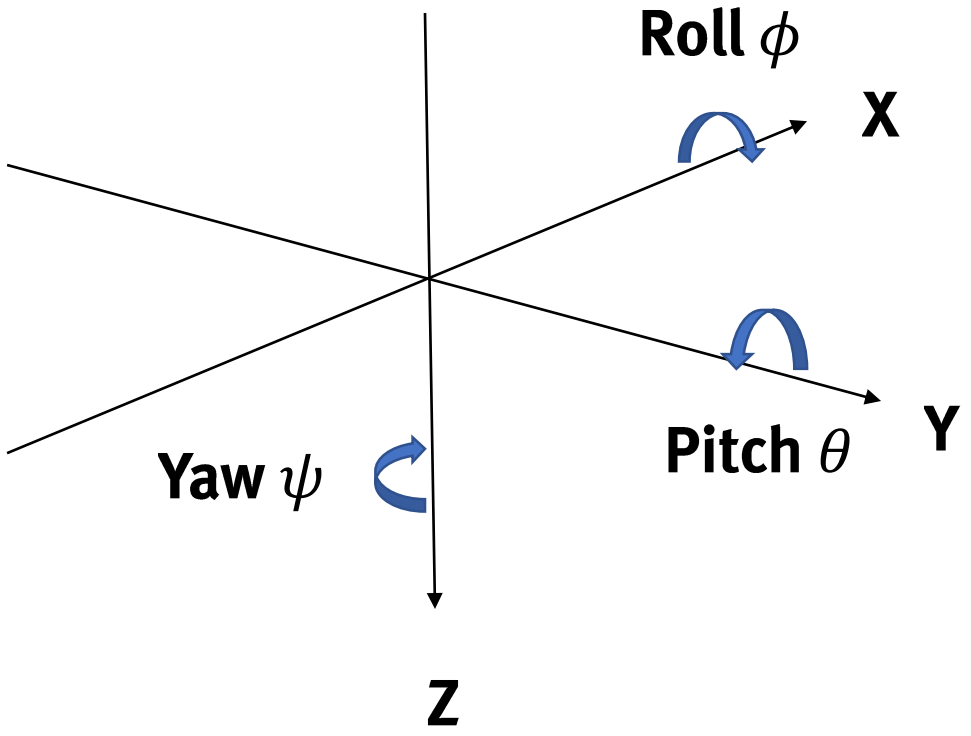


$$m = \left[m_x \; m_y \; m_z\right]^T$$

Goal

- Determine the Attitude (= orientation) of a system
- The Attitude is fixed to a given Frame (i.E. world frame)
- Can be given in different formats, for example:
  - Tait Bryan Angles (Roll, Pitch, Yaw)
  - Quaternions

Tait-Bryan Angles



Tait-Bryan Angles

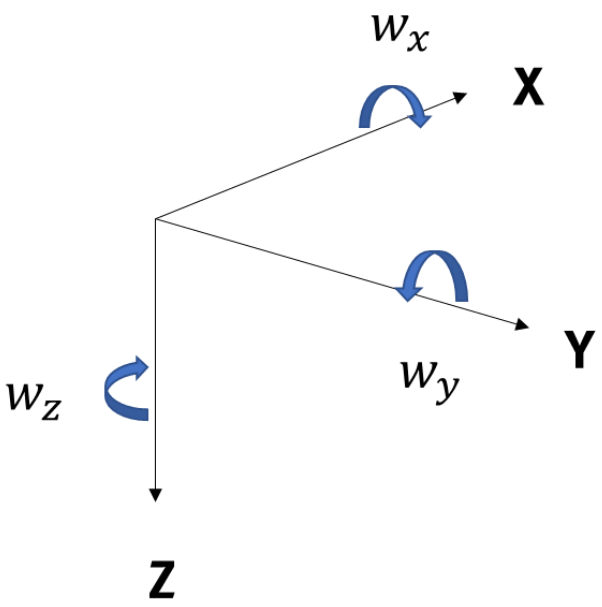
- Tait-Bryan Angles describe the orientation by 3 consecutive rotations in a fixed order (we will use Yaw – Pitch – Roll)
- Euler Angles (X-Y-X) $\neq$ Tait-Bryan Angles (i.E. Z-Y-X)
- Rotations around each axis can be described by rotation matrices

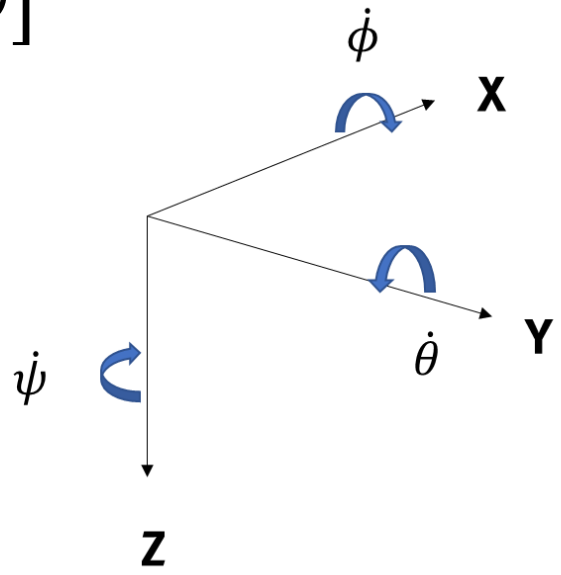
$$R = R_z(\psi) \cdot R_y(\theta) \cdot R_x(\phi)$$

Attitude Determination - Gyroscope

Gyroscope measures the angular velocity in 3 axis

$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T = \left[\dot{\phi} \;\; \dot{\theta} \;\; \dot{\psi}\right]^T$$





Attitude Determination - Gyroscope

Gyroscope measures the angular velocity in 3 axis

$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T = \left[\dot{\phi} \;\; \dot{\theta} \;\; \dot{\psi}\right]^T$$

→ We can use numerical Integration to calculate the attitude

But: We need to have an Initial State $[\phi \;\; \theta \;\; \psi]^T$

→ Use other sensor to calculate the initial State

Attitude Determination - Gyroscope

Gyroscope measures the angular velocity in 3 axis

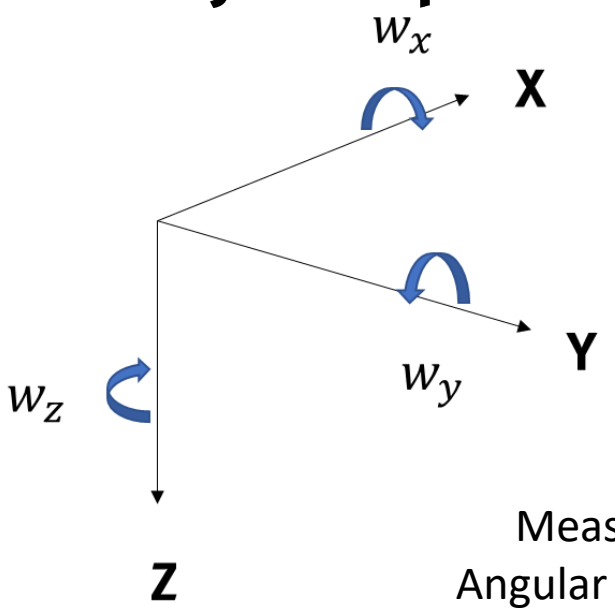
$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T = \left[\dot{\phi} \;\; \dot{\theta} \;\; \dot{\psi}\right]^T$$

→ Using numerical Integration with an initial State:

$$\begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_t = \begin{bmatrix} \phi \\ \theta \\ \psi \end{bmatrix}_{t-1} + \begin{bmatrix} \dot{\phi} \\ \dot{\theta} \\ \dot{\psi} \end{bmatrix}_t \cdot \Delta t$$

Attitude Determination - Gyroscope

Gyroscope measures the angular velocity in 3 axis:



$$\omega = \left[\omega_x \; \omega_y \; \omega_z\right]^T$$

$$\omega = \hat{\omega} + b_g + n_g$$

Measured Angular Velocity

Real Angular Velocity

Bias

White Noise

Attitude Determination - Gyroscope

Problems using <u>only</u> the Gyroscope:

- **Initial State needs to be calculated or assumed**
- **Bias will cause strong drift if not compensated (Integration)**
- **Noise will cause drift**

  → Only stable for short periods of time

Bias compensation

- Gyroscope Bias b_g adds an offset to the real value of the sensor

  → Bias needs to be removed to improve the solution

- Bias varies with external factors like temperature, but can be assumed stable after system start

  → Calculate Bias after start of the system

Bias compensation

Gyroscope measures the angular velocity in 3 axis:

$$\omega = \hat{\omega} + b_g + n_g$$

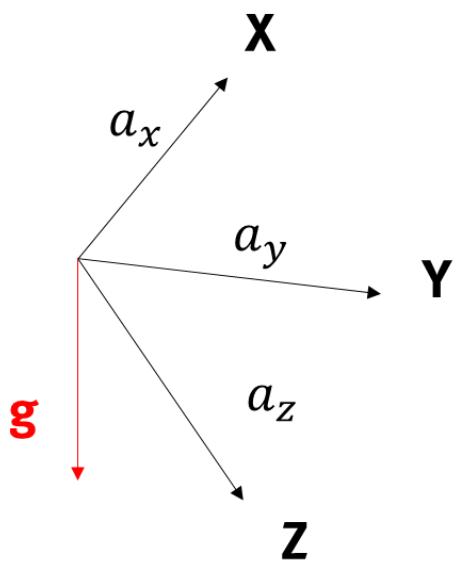
- **When system is at rest:** $\hat{\omega} = 0$
- **Averaging enough (> 100) values:** $\sum n_g = 0$

→ **Averaging over enough values when System is at rest:**

$$\omega = b_g$$

Attitude Determination - Accelerometer

Accelerometer measures the linear acceleration in 3 axis



$$a = \left[a_x \; a_y \; a_z\right]^T$$

Attitude Determination - Accelerometer

Accelerometer measures the linear acceleration in 3 axis

$$a = \left[a_x \; a_y \; a_z\right]^T$$

How can we can calculate the attitude using linear acceleration?

→ **Assuming we only rotate without acceleration**

→ **Use the gravity vector**

Attitude Determination - Accelerometer

Accelerometer measures the linear acceleration in 3 axis

$$a = \left[a_x \; a_y \; a_z\right]^T$$

→ Using the gravity vector to calculate the attitude

$$\phi_{acc} = atan2(a_y, a_z) \qquad \theta_{acc} = atan2\left(-a_x\,, \sqrt{a_y^2 + a_z^2}\right) \qquad \cancel{\gamma_{acc} = \mathrm{atan}\left(\frac{\sqrt{a_x^2 + a_y^2}}{a_z}\right)}$$

Attitude Determination - Accelerometer

Accelerometer measures the linear acceleration in 3 axis

$$a = \left[a_x \; a_y \; a_z\right]^T$$

→ Using the gravity vector to calculate the attitude

$$\phi_{acc} = atan2(a_y, a_z) \qquad \theta_{acc} = atan2\left(-a_x \, , \sqrt{a_y^2 + a_z^2}\right)$$

Attitude Determination - Accelerometer

Problems using <u>only</u> the Accelerometer:

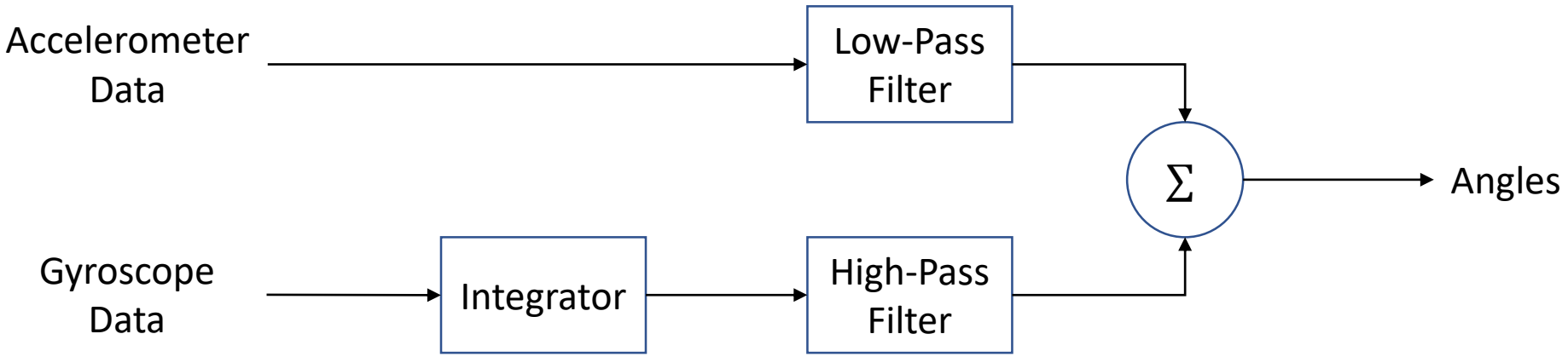
- **Only valid without external acceleration**
- **Measurements of translation and rotation gets mixed up**

  → Unstable for moving systems
- **Dependent on measuring the gravitational vector**

  → Can only be used for two axis (i.E. roll, pitch)

Complementary Filter

→ **Combine both sensors for better stability and accuracy**



Complementary Filter – A Problem

Using <u>only</u> Gyroscope and Accelerometer:

- **Only using one fixed vector (gravitational force)**

  → Yaw can not be calculated using the accelerometer

  → Drift will happen after some time

→ We need an additional sensor (Magnetometer)

Attitude Determination – Magnetometer

Magnetometer measures the magnetic field in 3 axis

$$m = \left[m_x \; m_y \; m_z\right]^T$$

How can we can calculate the attitude using magnetic field?

→ Use the earths magnetic field

Attitude Determination – Magnetometer

Magnetometer measures the magnetic field in 3 axis

$$m = \left[m_x \; m_y \; m_z\right]^T$$

→ Use the earths magnetic field as reference

But: We only want to calculate our attitude in the world frame

→ First rotate the measurement using the calculated tilt

Attitude Determination – Magnetometer

Rotate/Compensate the measurement by the current tilt

$$\widetilde{m} = \begin{bmatrix} \cos\theta & \sin\theta\sin\phi & \sin\theta\cos\phi \\ 0 & \cos\phi & -\sin\phi \\ -\sin\theta & \cos\theta\sin\phi & \cos\theta\cos\phi \end{bmatrix} \cdot \begin{bmatrix} m_x \\ m_y \\ m_z \end{bmatrix} = \begin{bmatrix} m_x\cos\theta + m_y\sin\theta\sin\phi + m_z\sin\theta\cos\phi \\ m_y\cos\phi - m_z\sin\phi \\ -m_x\sin\theta + m_y\cos\theta\sin\phi + m_z\cos\theta\cos\phi \end{bmatrix}$$

Calculate Yaw:

$$\psi = atan2\left(-\widetilde{m_y}, \widetilde{m_x}\right)$$

Attitude Determination – Magnetometer

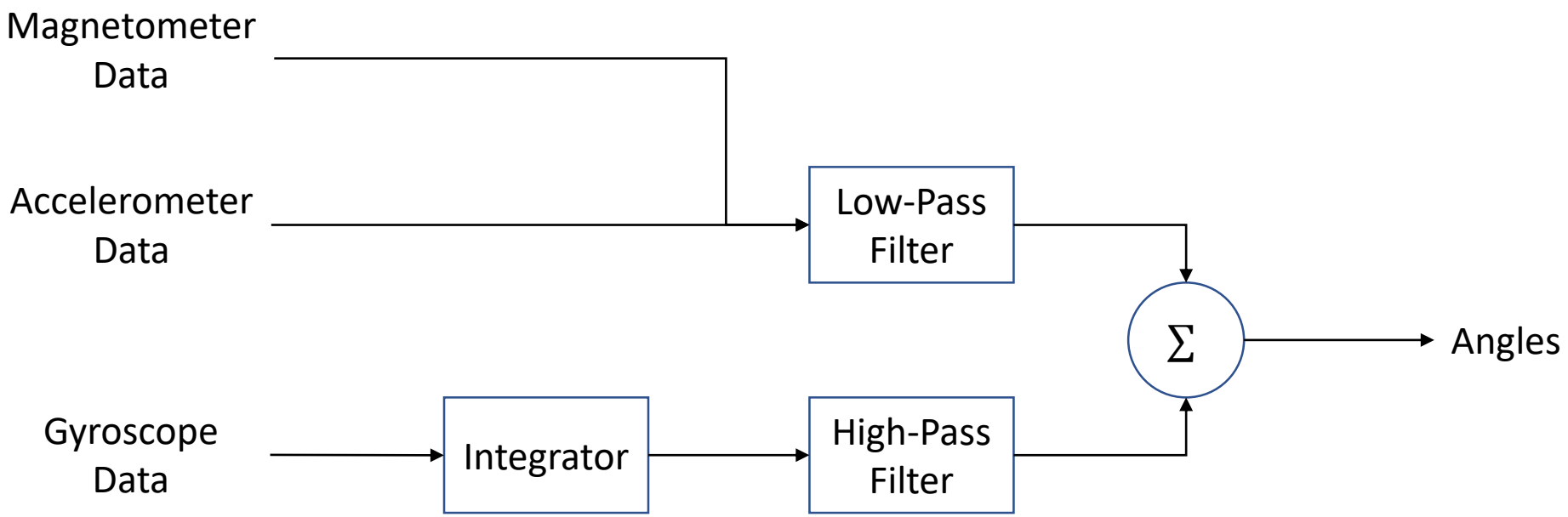
Problems using <u>only</u> the Magnetometer:

- **Only valid without external magnetic field**
- **Dependent on measuring the earths magnetic field**

→ Can only be used as additional sensor

Complementary Filter

→ **Combine all sensors for better stability and accuracy**



Low- and High-Pass Filters

Low Pass Filter

- **Used for Accelerometer and Magnetometer Data**
- **Removes High Frequency Noise**

$$\tilde{a}_{t+1} = (1 - \beta) \cdot a_{t+1} + \beta \cdot \tilde{a}_t$$

$\beta \in [0,1]$
$\tilde{a}$: *filtered accelerometer data*
a: *raw accelerometer measurement*

High Pass Filter

- **Used for Gyroscope Data**
- **Removes Low Frequency Noise**

$$\tilde{w}_{t+1} = (1 - \alpha) \cdot \tilde{w}_{t+1} + (1 - \alpha) \cdot (\tilde{w}_{t+1} - w_t)$$

$\alpha \in [0,1]$
$\tilde{w}$: *filtered gyroscope data*
w: *raw gyroscope measurement*

Data combination

- **Low- and High-Pass filters are combined when the data of the sensors is added**
- **Data combination for the complimentary filter:**

$$\begin{bmatrix}\phi\\ \theta\\ \psi\end{bmatrix}_{t+1} = \underbrace{(1-\beta)\cdot\left(\begin{bmatrix}\phi\\ \theta\\ \psi\end{bmatrix}_{t} + \begin{bmatrix}\phi_{gyro}\\ \theta_{gyro}\\ \psi_{gyro}\end{bmatrix}_{t+1}\right)}_{\text{High-Pass Filter for Gyroscope Angles}} + \underbrace{\beta\cdot\begin{bmatrix}\phi_{acc}\\ \theta_{acc}\\ \psi_{mag}\end{bmatrix}_{t+1}}_{\text{Low-Pass Filter for Acc/Mag Angles}}$$

Task1: Complementary Filter

Implement a complementary filter using Gyroscope and Accelerometer:

- Use the IMU data given by the thread from the previous tasks
- Calculate the initial State using the accelerometer and magnetometer
- Calculate the gyroscope bias values b_g at the start
- Correct the gyroscope measurements using the bias values
- Use the given formulas to calculate Angles from the sensor data
- Combine the values in the complementary filter
- Output the angles to view them in the GUI

Part 1: Questions

Please answer the following questions:

- **Why do we need multiple sensors to calculate our attitude?**
- **What's the difference between high & low pass filters, which one would you use for the gyroscope and why?**
- **Name one advantage and one disadvantage of the implemented simple complementary filter**

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